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System Requirements Specification

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Introduction

Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for a system that uses the Reachy Mini robot and AI-based movement detection to monitor and evaluate human movements during physiotherapy exercises at home. The project investigates whether the Reachy Mini can be used as an interactive and motivating exercise buddy. The system should be able to detect and potentially score common human movements and provide feedback or encouragement to the user. The SRS defines the functional and non-functional requirements of the proposed system and serves as a basis for its development and evaluation.

Scope

The scope of this project is to investigate the capabilities of the Reachy Mini for person detection and activity monitoring. The system will focus on detecting common movements such as walking, running, sitting, standing, and other relevant physiotherapy movements. The project will investigate whether existing AI and pose-estimation models can identify the movement being performed, determine whether it is performed correctly, and potentially provide a score. The Reachy Mini may also provide interactive feedback through movements, sounds, or voice to increase motivation. The project does not aim to solve a specific medical case or provide a complete medical monitoring solution. Instead, the focus is on investigating whether the Reachy Mini can identify and score general human movements and function as a motivating device for home-based physiotherapy.

Intended Users

The primary intended users are patients performing physiotherapy exercises at home. The system is intended to support users who perform exercises independently by providing motivation and potentially feedback on their movements. The Reachy Mini is intended to act as an interactive exercise buddy that can react to the user's movements and provide encouragement or feedback. The system may also be relevant to physiotherapists and healthcare professionals, who could use the Reachy Mini as a replacement assistant for appointments and scoring, to support the monitoring of patients' exercises. The original project document specifically mentions children with JDM (Juvenile Dermatomyositis) performing exercises that are assessed using the CMAS (Childhood Myositis Assessment Scale), but this project broadens the current project goal toward general physiotherapy movements and investigating the Reachy Mini as a motivating device.

System Overview

Product Description

The product is a movement detection system that uses the Reachy Mini robot's capabilities and AI algorithms to detect and score JDM patients movement during physiotherapy exercises at home. The system aims to provide motivation to patients. To support exercise performance.

Users and Stakeholders

The intended users of the Reachy Mini are JDM patients. But also, healthcare professionals who assess these patients. The robot could help assess the patients, without the patients needing to travel to the hospital as often.

Our client Maarten Vaessen who has tasked us with this project, is a senior researcher in AI for healthcare at Zuyd research center data intelligence (Zuyd University of Applied Sciences, n.d.).

Assumptions and Dependencies

The Reachy Mini can be programmed with the use of the Reachy Mini python SDK. Which is supported on Windows, MacOS and Linux. The python SDK requires python version 3.10-3.12 and the latest version of git, including git LFS (Huggingface, n.d.). It is assumed that the client will provide the Reachy Mini DIY kit.

Functional Requirements

The functional requirements describe what the system has to do. Each requirement has a unique identifier, is written using the modal verb *shall*, and applies to a single user in front of a single Reachy Mini during an exercise session.

Several requirements refer to *supported activities*. This means the closed set of movements the system is able to recognise: standing, sitting, walking, running, arm raise and sit-to-stand. The set is deliberately small, because the aim of the project is to find out whether recognition of this kind is feasible on the Reachy Mini, not to cover the full range of physiotherapy exercises.

Perception and local processing

Before an activity can be recognized, the system must detect and track the user. The activity recognition approach may either use an explicit skeletal or pose representation or infer activities directly from image or video data. Both approaches are therefore considered within the scope of this research. FR-02 additionally requires that this happens locally, which is the constraint that gives the project its research character and from which the privacy property in NFR-01 follows.

Table 1 – Perception and local processing functional requirements

ID	Title	Requirement
FR-01	Person Detection	The system shall detect a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.
FR-02	Local Processing	The system shall process captured camera data entirely on the Reachy Mini or the connected local computer, without requiring external processing for core activity-tracking functionality.
FR-03	Person Tracking	The system shall track the detected person continuously while the person remains visible within the camera's field of view during an exercise session.
FR-05	Pose Estimation	The system shall provide the input representation required by the selected activity recognition approach, which may be an explicit pose/skeletal representation or image/video data.
FR-18	User Detection Loss	The system shall detect that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.

Robot behaviour and interaction

The robot moves for two different reasons. In FR-04 the head moves in order to keep the user in view, which is a perception function. In FR-13 and FR-14 the head and antennas move in order to communicate with the user, which is an interaction function. How these two influence each other is one of the questions the project sets out to answer.

Table 2 – Robot behaviour and interaction functional requirements

ID	Title	Requirement

FR-04	User Following	The system shall move the Reachy Mini's head to follow the detected user while the user remains within the camera's field of view during an exercise session.
FR-12	Audio Feedback	The system shall provide audio feedback through the Reachy Mini's speaker whenever an audio feedback response is triggered.
FR-13	Head Movement Feedback	The system shall use movements of the Reachy Mini's head to provide feedback to the user whenever a head-movement feedback response is triggered.
FR-14	Antenna Movement Feedback	The system shall use movements of the Reachy Mini's antennas to provide feedback to the user whenever an antenna-movement feedback response is triggered.

Activity and exercise recognition

Three categories are distinguished because they place different demands on the recognition approach. Static positions can be classified from a single pose, dynamic movements require a sequence of poses over time, and physiotherapy exercises additionally have a defined correct execution.

Table 3 – Activity and exercise recognition functional requirements

ID	Title	Requirement
FR-06	Static Movement Recogntion	The system shall recognise all static positions listed as supported activities during an exercise session.
FR-07	Dynamic Movement Recognition	The system shall recognise all dynamic movements listed as supported activities during an exercise session.
FR-08	Physiotherapy Exercise Recognition	The system shall recognise all physiotherapy exercises listed as supported exercises during an exercise session.

Assessment and feedback

Recognising a movement is not enough on its own, since the robot is meant to act as an exercise buddy. FR-09 and FR-10 cover the judgement about the execution, and FR-11 covers the decision to respond. The assessment is explicitly non-clinical: the movement criteria describe geometric properties such as joint angles and symmetry, and do not constitute a medical score.

Table 4 – Assesment and feedback functional requirements

ID	Title	Requirement
FR-09	Movement Assessment	The system shall assess the execution of supported physiotherapy exercises using the predefined movement criteria for each exercise.
FR-10	Repetition Detection	The system shall detect completed repetitions according to the predefined repetition criteria during an exercise session.

Session management

An exercise session is the unit of operation of the system: the user starts it, the system observes continuously during it, and the results are reported at the end.

Table 5 – Session management functional requirements

ID	Title	Requirement
FR-11	Session Management	The system shall allow the user to start and end an exercise session.
FR-12	Exercise Monitoring	The system shall monitor the user's movements continuously throughout an active exercise session.
FR-13	Exercise Result	The system shall provide the user with the detected exercise results and assessment results at the end of an exercise session.

Non-Functional Requirements

The non-functional requirements describe the qualities the system has to exhibit rather than the functions it has to perform. Each is expressed as a measurable target, so that the outcome of the investigation can be reported objectively.

Privacy is a driving constraint rather than a secondary concern, because the system observes a patient exercising at home. Accuracy determines whether the system is useful at all, since feedback based on a misrecognised movement is worse than no feedback; the threshold of 80% demonstrates technical feasibility within the scope of this project and is not a level that would be sufficient for clinical use. Reliability and usability follow from the fact that the system is operated by a patient without technical support present, and safety follows from the robot moving in the physical presence of a user who may be a child.

Table 6 – Non-Functional Requirements

ID	Title	Requirement
NFR-01	Privacy	The system shall ensure that captured video and session data remain on the Reachy Mini and/or the connected local computer for the entire duration of system operation and shall not be transmitted to external systems.
NFR-02	Recognition Accuracy	The system shall provide activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance shall be measured and reported.
NFR-03	Reliability	The system shall complete an exercise session without requiring an application restart under normal operating conditions.
NFR-04	Usability	The system shall allow first-time users to start an exercise session without assistance from the development team.
NFR-05	Safety	The system shall not perform unexpected or uncontrolled robot movements during normal operation.

Bibliography

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